

Scalability & Future Plan

Date	07 July 2024
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The application currently supports a limited number of concurrent users.	Performance may decrease under heavy traffic.	High
2	The model is trained on a single dataset and may not generalize to all real-world scenarios.	Prediction accuracy may vary for unseen data.	Medium
3	Basic authentication and security features are not implemented.	Reduced security for production deployment.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on the Render free tier.	Deploy on a higher-capacity cloud server with load balancing.
2	Data Storage	Uses local dataset and model files.	Store data and models in a cloud database or object storage.
3	Performance	Single Flask application handles all requests.	Use a production WSGI server (e.g., Gunicorn) with multiple workers and optimize model inference.
4	Security	Basic Flask application without user authentication.	Implement user authentication, HTTPS, and input validation for improved security.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Improve the prediction model using larger and more diverse datasets.	Next 3 Months	Increased prediction accuracy and reliability.
Phase 3	Add user authentication, dashboards, and prediction history.	Next 6 Months	Enhanced user experience and secure access.
Phase 4	Deploy the application on a scalable cloud platform with continuous monitoring and automated updates.	Next 12 Months	Improved scalability, availability, and production readiness.